[1] Department of Meteorology, University of Reading;
[2] National Centre for Atmospheric Science; [3] UK Met Office

Monsoons at the Kilometre Scale

Jamie Todd [1] *(Supervisors: Andrew Turner [1,2], Akshay Deoras [1,2], Eliza Karlowska [1,2], Richard Jones [3])*

Introduction & project overview:

Monsoons are a critical water source for over two-thirds of the world's population, but their **representation** in global **climate models suffers due to coarse horizontal resolution** (Δx) and treatment of sub grid-scale processes. Km-scale (⩽ 10 km Δx) simulations better represent processes e.g. **diurnal cycle** of convection, mesoscale circulations key to accurately representing monsoons. Project investigates:

- Simulated **monsoon behaviour** in India & West Africa at a **range of Δx**, and with various science settings.
- **Global simulations** enable assessment of monsoon **impact** on the **large-scale circulation**.

Data: suite of **year-long** (2020-21), global, atmosphere- (and land-) only simulations run by the Met Office as part of the UK's contribution to the **DYAMOND-3** initiative.

Model	RAL3	$CoMA9_{5km/10km}$	$GAL9_{10km/60km}$
Native Δx	~5 km (N2560)	~5 km (N2560) & ~10 km (N1280)	~10 km (N1280) & ~60 km (N216)
Convection	Explicit convection	New mass-flux; CoMorph Trailblazer (scale-aware)	Gregory-Rowntree 6A (close to operational configuration)
Microphysics	CASIM (double moment)	Wilson & Ballard (single moment)	Wilson & Ballard (single moment)
Large-scale cloud	Bimodal – diagnostic	PC2 – prognostic	PC2 - prognostic
Boundary-Layer	Blended 1D (Lock) and 3D (Smagorinsky)	Unblended 1D	Unblended 1D

Research aims:

1*:* Characterize general monsoon behaviour e.g. **diurnal cycle, scale-interactions** with local circulations and larger scales.

2*:* Investigate the relationship between monsoons & **larger-scale modes of variability** e.g. MJO, BSISO and whether this changes at km-scale.

3*:* Test the impact of orography, land surface state, and their feedbacks on the large-scale circulation using **sensitivity experiments**.

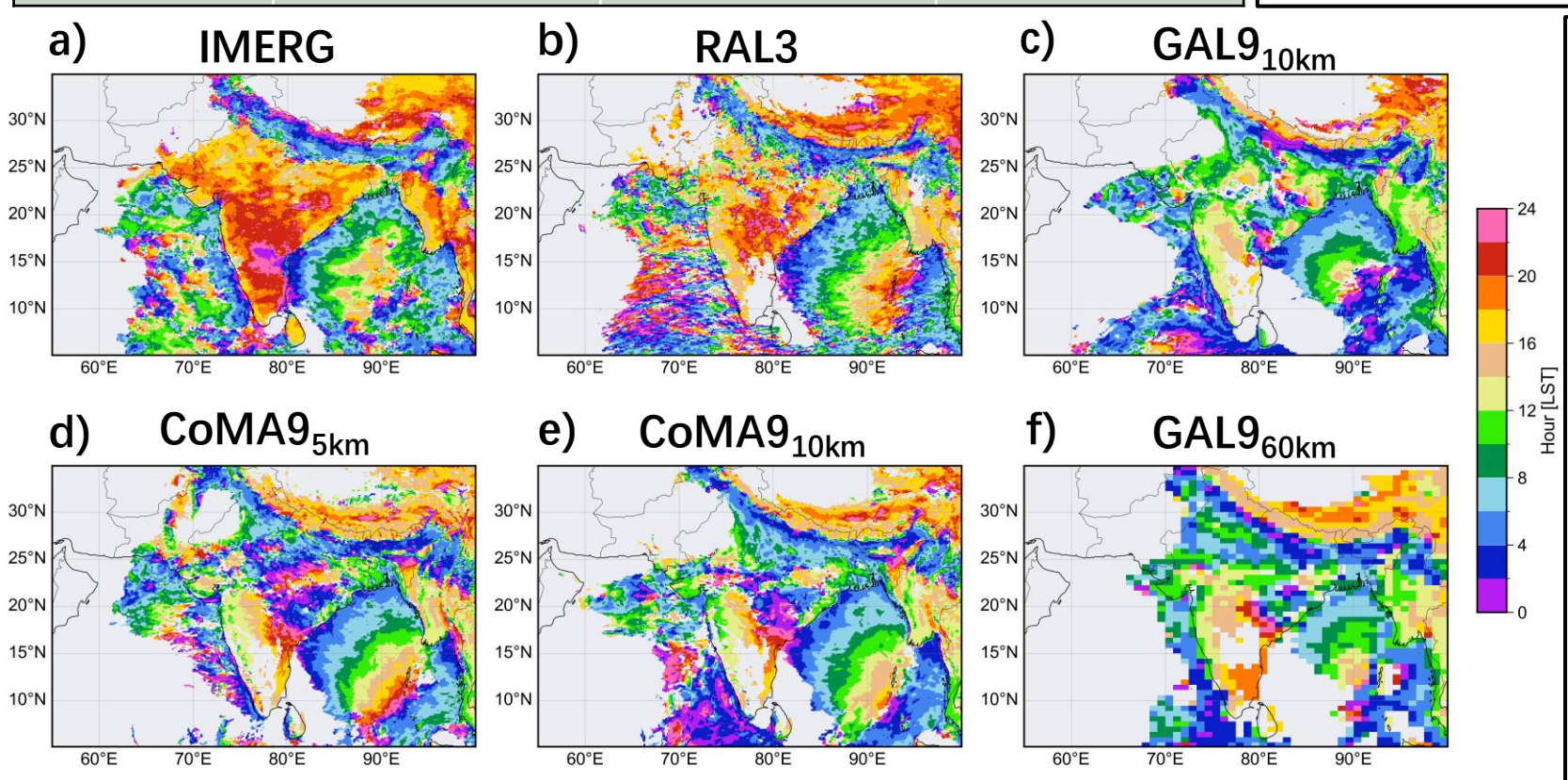


Fig. 1. *LST of precipitation maximum computed as first harmonic of diurnal cycle in (a) GPM-IMERG observations and (b-f) model simulations (RAL3, $GAL9_{10km}$, $CoMA9_{5km}$, $CoMA9_{10km}$ & $GAL9_{60km}$) for JJAS over India. Data, except $GAL9_{60km}$, were regridded to 0.1° resolution.*

- **RAL3 better matches IMERG** in the timing of peak rainfall over land than other simulations.
- Differing structure of offshore-propagating convection from East coast of India in early morning.